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Research on Classification of Chinese Herbal Medicine Pieces Based on Fusion of YOLOv8n and HSV

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***Abstract*—The quality of Chinese herbal medicine pieces, as the core substance in TCM clinical prescriptions, directly determines the efficacy and safety of clinical treatment and serves as a pivotal control point in the standardization process of traditional Chinese medicine. This study presents a novel color grading framework combining the HSV color space model with the YOLOv8n object detection algorithm. HSV enables accurate separation of hue, saturation, and brightness components, reflecting color characteristics such as dryness, mildew, and vividness. YOLOv8n assists in locating the main herbal area, effectively removing background noise and enabling precise region-based color analysis. The proposed model was evaluated on a dataset of 3,879 images covering 201 herbal types, showing a high precision of 93.62% and recall of 96.23%. Experiments demonstrate that this HSV-YOLOv8n integrated model outperforms baseline algorithms, especially in detecting surface changes and identifying moldy regions. It achieves reliable classification even in visually complex or non-standard scenarios. The framework shows strong potential for real-time deployment in automated production environments, promoting the digital standardization of Chinese herbal medicine quality.**

Keywords: Chinese herbal medicine, color grading, YOLOv8n, HSV, image recognition, quality evaluation

I. Introduction

As the core raw material in clinical TCM and the production of Chinese patent medicines, the quality of Chinese herbal medicines directly affects the efficacy and therapeutic effects. Color, as a key indicator in the sensory quality evaluation of Chinese herbal medicines, has a high discriminant value, especially in rhizome Chinese herbal medicines. Traditional manual grading methods rely on empirical judgment, which is highly subjective and inefficient, and it is difficult to meet the needs of the modern Chinese medicine industry for standardization, automation and high-throughput grading. In recent years, with the development of image recognition technology and color space modeling theory, the use of digital images for color grading of herbal medicines has become a research hotspot, especially the HSV color model, which is widely used in color analysis tasks due to its consistency with the color perception of the human eye.

Chen et al. (2021) explored the application prospects of artificial intelligence in TCM tongue diagnosis image recognition, providing technical support for the digitization of traditional medicine [1] . Chen et al. (2023) further proposed an automatic tongue image analysis algorithm for TCM diagnosis, proving the effectiveness of combining color space with feature extraction algorithm [2] . Liu et al. (2023) realized multi-scale image enhancement in RGB and HSV space, revealing the advantages of HSV channel in color correction [3] . Zhou et al. (2024) proposed the IACC method for image color correction under mixed lighting conditions, emphasizing the important role of color space in complex scene perception [4] . Wang et al. (2023) improved the robustness of images under color shift and scattering interference based on the maximum likelihood adaptive color correction algorithm [5] .

Therefore, this paper focuses on the core goal of "color grading of Chinese herbal medicines based on HSV", combines the YOLOv8n target detection algorithm to locate the herbal medicine area, and establishes a grading process suitable for color feature extraction of herbal medicine images, aiming to provide technical support and method reference for quality standardization and intelligent identification of Chinese herbal medicines.

II. HSV Color Space Model and Color Feature Extraction Theory of Chinese Herbal Medicine Pieces

A. HSV color model principle

The HSV (Hue, Saturation, Value) color model is a color representation method based on human perception mechanisms and is widely used in image processing, object recognition, and color analysis. The model converts color from the RGB representation in a rectangular coordinate system to a conical coordinate system that is more consistent with visual cognition. As shown in Figure 1 , the HSV model is usually geometrically represented as a cone, in which the threedimensional attributes of color are described by hue (H), saturation (S), and brightness (V) [6] .

Hue indicates the type of color, expressed as a circular angle ranging from 0° to 360°. For example, 0° corresponds to red, 120° corresponds to green, and 240° corresponds to blue. Each hue is evenly distributed on the hue circle according to the angle. Saturation indicates the purity of the color, ranging from 0 to 1. When the S value is 0, the color appears gray, and when the S value is 1, the color is the most saturated. Brightness describes the lightness and darkness of the color, and the value range is also 0 to 1. The larger the V value, the brighter the color. When V is 0, all hues tend to black [7] .

In the application of color grading of Chinese herbal medicines, the HSV model can effectively separate color information from light intensity and is suitable for extracting stable and anti-interference color features. Especially when comparing color differences and identifying moldy areas, the H channel can reflect the hue drift trend, while the S and V channels can quantify the color vividness and brightness changes, providing an important parameter basis for color grading.

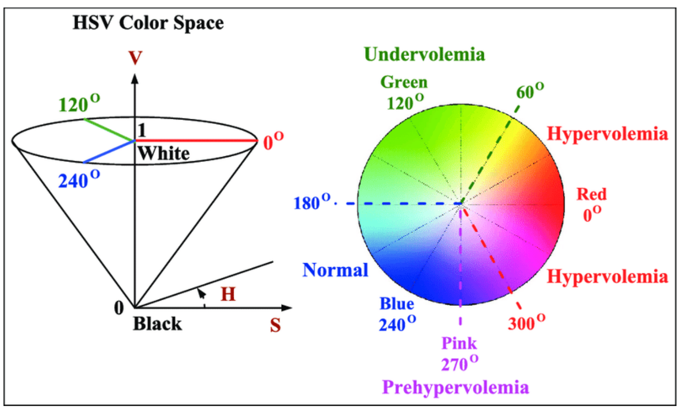

Figure 1. Principle of HSV color model

B. *Analysis of the Demand for Color Grading of Chinese Herbal Medicines*

The color of Chinese herbal medicines is one of the important indicators of their sensory quality, especially for rhizomes. The color uniformity, saturation and brightness are of great reference value for judging their degree of dryness, mildew and storage quality. Traditional manual grading relies on visual experience and is difficult to maintain consistency among large batches of medicinal pieces. Root and rhizome Chinese medicinal materials such as turmeric, white peony root, and salvia miltiorrhiza show significant color differences in different origins and after different processing, mainly manifested in the transition and depth changes between yellow, brown, reddish brown and other color systems. For example, high-quality slices of Scutellaria baicalensis are bright yellow, while inferior pieces tend to be dark brown or gray [8] .

Therefore, it is necessary to establish a standard color grading mechanism based on visually perceptible color changes, and combine image processing technology to achieve objective judgment. The grading strategy is usually based on the ability of the H channel (hue) in the HSV model to distinguish color types, combined with the characteristics of the S channel (saturation) reflecting color purity and the V channel (brightness) reflecting the degree of lightness and darkness, to divide the color of the medicinal pieces into bright yellow, light yellow, dark yellow, brown and other levels, providing a basis for the discrimination rules for the establishment of the subsequent automated grading system.

C. *Modeling method of HSV features of medicinal pieces*

In order to realize the automatic grading of the color of Chinese herbal medicines, it is necessary to perform feature modeling on the image data of the three HSV channels. First, in the target area detected by the YOLOv8n model, the H, S, and V channel histograms of the image pixels are constructed, and the distribution characteristics of each channel are counted. With the H channel as the core, its mode is calculated as the main color value of the herbal medicine, which is defined as follows:

$$H_{\text{mode}} = \arg \max_{h \in \left[0^0, 360^\circ\right]} freq(h) \tag{1}$$

Where $freq(h)$ is the frequency of a certain H value appearing in the image. The mean of the S channel is used to measure the purity of the color and is expressed as:

$$\overline{S} = \frac{1}{N}\sum_{i=1}^{N} S_i \tag{2}$$

V The channel average $\overline{V}$ represents the image brightness and is used to determine the glossiness and dryness of the medicinal material surface. Based on a large number of sample statistical analyses, the typical threshold range of each channel is defined. For example, bright yellow medicinal materials often have $H \in \left[40^\circ, 60^\circ\right], \overline{S} > 0.6, \overline{V} > 0.7$ the following characteristics [9] .

Based on the above parameters, a color grading judgment rule set is constructed, and the color grade of medicinal pieces is output by weighted combination or decision tree method, so as to realize automatic grading and recognition of batch images of medicinal pieces and provide real-time and stability support for production line deployment.

D. *Theoretical basis of mold color recognition in HSV color space*

Mildew is an important indicator of the deterioration of the quality of Chinese herbal medicines. Its color characteristics are usually manifested as epidermal hue shift, brightness decrease, and uneven saturation. The HSV model can effectively separate these characteristic changes, thereby realizing the identification of moldy areas. The hue H channel is sensitive to moldy areas, and often drifts from the original yellow to green or gray-blue. The V channel value decreases significantly, reflecting the weakening of brightness, while the S channel shows discrete fluctuations in color purity.

The mildew detection process includes extracting the three HSV channels and calculating the deviation degree of each pixel point based on the color range of the normal sample.

Taking hue shift as an example, the mildew discrimination function is defined as:

$$\Delta H_i = \left| H_i - H_{ref} \right| \qquad \Delta H_i > \theta_H \text{ Marked as abnormal} \quad (3)$$

Where H_i is the current pixel hue value, H_{ref} is the normal hue average of this type of medicinal material, θ_H and is the set threshold (such as 15°). The V channel and S channel are similar, and the threshold is used θ_V, θ_S to determine the dark spot or color blurred area respectively .

In the actual system, the contour of the abnormal area is extracted by combining the region growing algorithm, and its area ratio is counted to construct four levels of mold grade (normal, mild, moderate, and severe), and finally the availability judgment result of the medicinal material is output.

III. Design of the Extraction Method of the Decoction Area Based on YOLOv8n

A. Necessity analysis of region positioning before color grading

In the automated grading system of Chinese herbal medicines, color is a key feature that directly affects the accuracy and stability of grading. However, images of herbal medicines often contain cluttered backgrounds, uneven lighting, and differences in color distribution in different areas, which makes it challenging to directly extract color features from the overall image. Therefore, accurate regional positioning becomes the primary step to ensure the accuracy of subsequent color grading. The goal of regional positioning is to accurately select the main area of the herbal medicine, exclude irrelevant background parts, and reduce the interference of color difference, shadows, and noise on color analysis.

The use of the YOLOv8n target detection algorithm can effectively solve this problem. Through the target detection function of the deep learning model, YOLOv8n can identify the target area in a complex background and accurately assign a bounding box to each herbal medicine image. This process will greatly improve the accuracy of subsequent HSV color feature extraction, avoid unnecessary background information interference, and thus improve the performance of the color grading system. In practical applications, YOLOv8n can locate and classify targets in herbal medicine images in real time through its efficient convolutional neural network (CNN) structure, achieving efficient image segmentation and feature extraction. With the help of target positioning, the system can calculate the H, S, and V values in a specific area, thereby providing reliable input data for subsequent color grading, and ultimately achieving automated and standardized quality assessment of Chinese herbal medicines.

B. YOLOv8n algorithm structure and improvement strategy

YOLOv8n (You Only Look Once Version 8n) is an efficient target detection network. It is based on an improved convolutional neural network (CNN) structure and improves the ability to accurately identify target areas (such as edges of Chinese herbal medicines, color areas, etc.) in Chinese herbal medicine images through optimized network layer design. As shown in Figure 2, its architecture consists of three modules: Backbone, Neck, and Head. The design of each module ensures high-precision detection while improving the model reasoning speed.

In the Backbone module, multiple convolutional layers and C2f (Channel-to-Feature) structures are used to extract features and integrate multi-scale information. The C2f module uses multiple convolutional layers for feature separation and fusion, as shown in Figure 2. Multiple bottleneck layers further extract effective features by reducing channels and increasing the amount of calculation. At the same time, the SPPF (Spatial Pyramid Pooling Fast) module in the Neck part optimizes the spatial pooling operation and enhances the robustness of the model to targets of different scales through multiple pooling layers (MaxPool2d), especially in the positioning of medicinal pieces in complex backgrounds.

box regression and classification label through fine branch detection and classification structure, and optimizes the performance of YOLOv8n in target detection. Mathematically, the regression of the bounding box can be expressed as:

$$\text{Box Loss} = \lambda_1 \cdot \sum_i \left(x_i - \hat{x}_i\right)^2 + \lambda_2 \cdot \sum_i \left(y_i - \hat{y}_i\right)^2 + \lambda_3 \cdot \sum_i \left(w_i - \hat{w}_i\right)^2 + \lambda_4 \cdot \sum_i \left(h_i - \hat{h}_i\right)^2 \quad (4)$$

Among them, x, y, w, h are the center coordinates and width and height of the target frame, $\hat{x}, \hat{y}, \hat{w}, \hat{h}$ are the predicted values, $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ and are the weight coefficients. Combined with the multi-layer feature fusion and efficient detection module of YOLOv8n, it can accurately locate the target of Chinese herbal medicine pieces in the image of Chinese herbal medicine pieces, and provide high-quality regional input data for subsequent color grading.

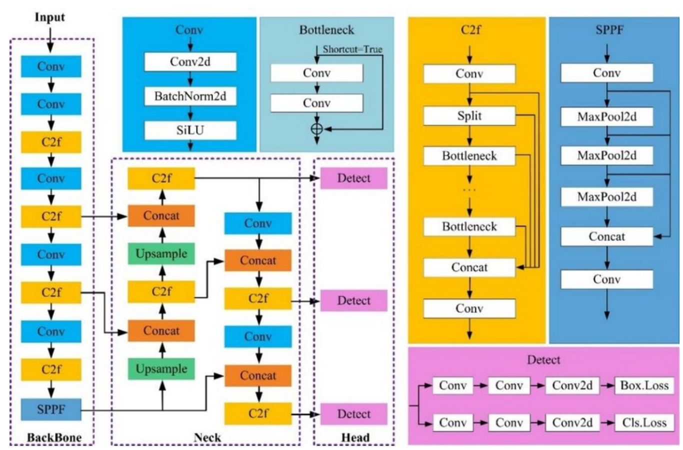

Figure 2. YOLOv8n algorithm structure

C. YOLO-assisted HSV color extraction process design

Based on the YOLOv8n target detection results, a color feature extraction module is constructed to improve the accuracy and efficiency of color grading in TCM slice images. First, the YOLOv8n network accurately locates the target area, i.e., the main part of the slice, through its Backbone, Neck, and Head modules, and generates a bounding box. This framed area effectively eliminates the interference of background clutter and uneven lighting, and provides a clear and unified regional input for subsequent color analysis.

the selected area , the HSV color space conversion is first performed to convert the image from RGB space to HSV space. This process can effectively remove the impact of lighting changes on color, focusing on the changes in the H channel (hue), S channel (saturation) and V channel (brightness). For each medicinal material target, the pixel values of the three channels H, S, and V in the area are extracted, and the average hue value, saturation mean, and brightness value are calculated. In particular, the hue (H) can be calculated by the following formula to calculate its average value:

$$H_{avg} = \frac{1}{N}\sum_{i=1}^{N} H_i \tag{5}$$

Among them, N is the number of pixels in the area, H_i is the hue value of the i - th pixel. Similarly, the average values of saturation (S) and brightness (V) can also be calculated by the corresponding formula. In order to further quantify the purity and brightness of the color, a color scoring index is designed to judge the color quality and grade of the medicinal pieces. The score is based on the stability of the H channel (such as the degree of deviation from the normal hue), the mean of the S and V channels, and comprehensively considers the vividness and brightness of the color.

IV. HSV Color Grading Experiment and Result Analysis

A. Dataset construction and preprocessing

In order to effectively evaluate the recognition performance of the classification model of Chinese herbal medicine pieces that integrates YOLOv8n and HSV features, this study constructed a dedicated dataset covering 201 kinds of Chinese herbal medicine pieces and a total of 3879 images. All images were collected by the outpatient pharmacy of the Institute of Chinese Materia Medica, China Academy of Chinese Medical Sciences, and were taken under standardized conditions using a Canon EOS 80D camera with a uniform white background to ensure that the image background is clean and the features are prominent.

The image resolution is uniformly adjusted to adapt to the input requirements of the YOLO series models. On this basis, the images are also subjected to a series of data enhancement processing, including noise filtering, image rotation, translation, brightness enhancement, etc., which significantly improves the diversity and robustness of the samples. These operations improve the model's learning ability under different postures, angles and lighting conditions without changing the original semantic information.

The dataset is divided into training set, validation set and test set in the ratio of 7:2:1, and further divided into six categories: root and stem segments (category A), root and stem slices (category B), fruit or seeds (category C), skin (category D), leaves (category E) and others (category F). Some sample images contain multiple medicinal pieces instances, with high morphological complexity and arrangement freedom, which enhances the generalization training effect of the model in multi-instance scenarios.

In addition, in order to examine the model's adaptability to non-standard shooting conditions, external medicinal piece image data crawled from public channels on the Internet is also introduced as an additional test set. These images have different shooting angles and complex backgrounds, which provide a multi-dimensional verification scenario for model performance evaluation and can also test its generalization stability and robustness in practical applications.

B. Overall Model Performance Evaluation

To comprehensively evaluate the classification performance of the proposed YOLOv8-TCM model, several mainstream object detection algorithms were selected for comparison, including SSD, Fast R-CNN, and a range of models from the YOLO series. As shown in Table 1, the comparison metrics include Precision, Recall, mean Average Precision at IoU thresholds from 0.5 to 0.95 (mAP@50–95), and model parameter size.

The YOLOv8-TCM model outperformed all baseline models across most metrics. Specifically, it achieved a precision of 93.62%, a recall of 96.23%, and a mAP@50–95 of 84.16%, indicating strong overall detection capability. While the YOLOv8s model had a marginally higher recall (96.78%), it required four times more parameters (11.2 million) compared to YOLOv8-TCM (2.8 million), significantly limiting its deployment potential on resource-constrained edge devices.

Compared to the base YOLOv8n model, the proposed model improved mAP@50–95 by 4.39% while reducing parameter size by approximately 15%, further validating the efficacy of integrating HSV color features and the lightweight CA + DySnakeC2f module fusion strategy. The details are shown in Table 2 and Figure 3:

Table 1 Model Performance Comparison

Model	Precision (%)	Recall (%)	mAP@50-95 (%)	Params ($\times 10^6$)
SSD	79.73	86.36	76.55	13.3
Fast R-CNN	87.42	89.03	80.56	17.1
YOLOv5n	84.42	88.81	78.18	3
YOLOv6n	80.86	85.92	76.81	4.8
YOLOv6s	86.89	90.55	78.95	16.7
YOLOv8n	88.41	89.95	79.77	3.4
YOLOv8s	92.26	96.78	82.72	11.2
YOLOv10n	90.24	92.02	81.85	3.1
YOLOv8-TCM	93.62	96.23	84.16	2.8

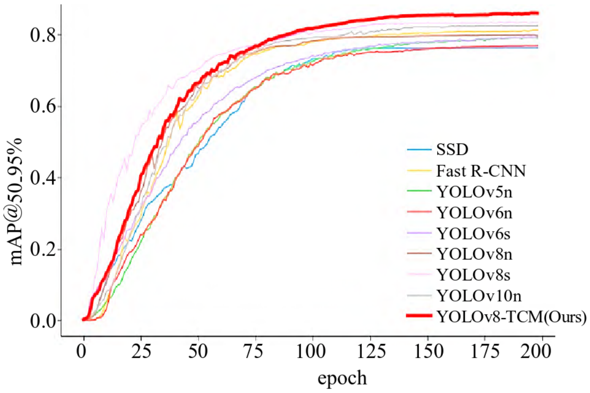

Figure 3. Change curve of model training process

The superiority of YOLOv8-TCM lies not only in its high accuracy but also in its compact structure, which allows for real-time processing and easier integration into practical Chinese herbal medicine classification systems.

C. *Accuracy Analysis for Different Herbal Categories*

To evaluate the classification performance of YOLOv8-TCM across various types of Chinese herbal medicine pieces, per-category accuracy was calculated for each of the 201 classes. As shown in Table 2 (Sample) and Figure 4, the vast majority of categories achieved high recognition accuracy, with 75% of all classes exceeding 90% accuracy, demonstrating the model's strong generalization ability across diverse herbal forms.

Skin-type herbs, such as Chuanmuxiangpi and Hehuanpi, demonstrated the highest classification accuracy, benefiting from their regular shapes (e.g., rectangular or square) and smooth surfaces. For instance, Baixianpi achieved 98.7%, and Chenpi, Guanhuangbo, and Hehuanpi were all correctly classified at 100% accuracy.

Root and stem slice types also showed consistently strong results due to their distinct internal texture and consistent cutting patterns. For example, Ganjiang reached 92.5%, Daxueteng 98.5%, and Haifengteng 96.2%.

Fruits and seeds, typically round or ovoid with vivid colors, yielded high accuracy rates as well. Lianqiao, Jinyingzi, Gouqi, and Doukou all exceeded 94% accuracy.

Conversely, leaf-type herbs posed greater challenges due to their irregular, thin, and sometimes curled morphology. The recognition accuracy of Xiakucao was only 68.4%, while Shenjincao and Sangye achieved around 85%. This variability underscores the difficulty in modeling highly deformable, low-texture targets.

Notably, herbs like Weilingxian, Juluo, and Nüzhenzi achieved only 70% accuracy, largely due to fragmentation, severe occlusion, and shadow interference, as well as insufficient sample diversity. For example, Nüzhenzi had only eight samples, most of which lacked varied placements, leading to overfitting and weakened feature learning.

Table 2 (Sample): Accuracy of medicinal material identification

Herb Piece	Accuracy (%)
Xiangru	97.5
Zhi Huangqi	97.4
Qingfengteng	96.9
Maidong	92.7
Chishao	87.6
Zexie	91.6
Xiakucao	68.4
Shancigu	96.3
Baishao	94.3
Huangqi	97.8
Honghua	96.7
Digupi	89.3
Jiaomaiya	92.1
Chuanxiong	99.1
Danshen	93.4
Zhuru	95.3
Shudihuang	96
Jinyinhua	88.1
Huanglian	97.6
Tianma	98.6

These findings highlight that while the model performs well in most structured categories, its accuracy drops for fragmented, overlapping, or under-represented herbs. This emphasizes the need for targeted improvements in data augmentation, sample balancing, and model robustness strategies in future research.

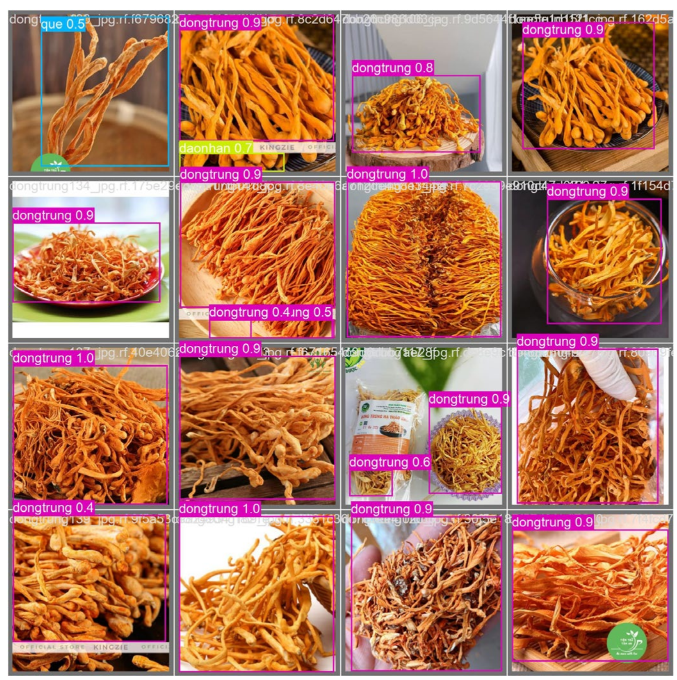

Figure 4. Visual verification of medicinal material identification

V. Conclusion and Outlook

This study develops an intelligent color grading system for Chinese herbal medicine using HSV color modeling and YOLOv8n-based region detection. By leveraging the visual consistency of HSV color space and the robust localization capability of YOLOv8n, the system accurately identifies medicinal regions and extracts reliable color features for grading and quality assessment. The model achieves outstanding performance across most herbal categories, with more than 75% surpassing 90% classification accuracy. In addition, it maintains robustness against complex backgrounds, uneven lighting, and sample variability. While leaf-based and fragmented herbs remain challenging due to irregular morphology, the model still provides stable results and highlights the importance of improved data diversity. The lightweight design of the system supports fast inference and real-time application, making it feasible for industrial integration. Overall, this research offers a scalable, automated solution for standardized herbal grading and contributes valuable insights into the digital transformation of traditional Chinese medicine quality control.

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